

The Accumulating-History Extraction

A Real Developmental Test: Cortical Neurogenesis

*The regime Application 02 pointed to · the retention coordinate of
the developing mouse neocortex · the interface bridge screened
against the measured leaving fraction · an honest failure that
grazes the fixed point*

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1. THE REGIME THIS TEST REQUIRES

Application 02 screened cortical avalanches in their stationary self-organised state and rejected them: in that regime the network is almost entirely quiescent between cascades, retention $x \rightarrow 1$ and continuation pressure $\chi \rightarrow 1$, and $\chi \rightarrow 1$ is merely the branching ratio $\sigma \rightarrow 1$ restated. The diagnosis named the regime the bridge actually needs — an accumulating processing history, in which the terminal class grows monotonically as stellar oxygen does over the burning epoch, rather than an instantaneous fraction of a recycling steady state. Neuronal development is exactly such a history: progenitors proliferate while neurons accumulate irreversibly. This note carries out the extraction there, on measured data, with the rule frozen in advance.

2. THE DATA

The developing mouse neocortex has a single, classical, quantitatively measured control variable: the leaving fraction Q , the proportion of a progenitor's daughters that exit the cell cycle and become neurons, measured across the eleven cell cycles of the neuronogenetic interval E_{11} – E_{17} by Takahashi, Nowakowski and Caviness (Journal of Neuroscience, 1996). Q rises monotonically from about 0.08 at E_{11} through 0.5 near E_{14} – E_{15} to unity at E_{17} ; the cell-cycle length T_c lengthens from roughly nine to eighteen hours across the same interval. These are the measured record and nothing here is fitted to them.

RECORD — MEASURED INPUT — Q per embryonic day and T_c per embryonic day are taken from Takahashi, Nowakowski & Caviness (1996), Tables 2–3, with the endpoints E_{11} and E_{17} extrapolated from the paper's stated cycle-1 and cycle-11 values. The trajectory below is integrated cell cycle by cell cycle from these numbers alone.

3. THE EXTRACTION

The seven-part machine is the proliferative epithelium: the progenitor pool is the retained class, the accumulating neurons the terminal class, activation of the differentiation programme the one-way conversion, and the cell cycle the clock. Per cycle the progenitor count updates as $A \rightarrow 2(1-Q)A$ and the neuron count as $B \rightarrow B + 2Q \cdot A$; production and conversion are the gross births into each class, $P = 2(1-Q)A$ and $K = 2Q \cdot A$. The retention coordinate is $x = A/(A+B)$ and the continuation pressure $\chi = K/(P \cdot x) = [Q/(1-Q)]/x$, with $\beta = 0$ for the closed epithelium. Integrating the measured $Q(E)$ from E_{11} gives the trajectory below: as neurons accumulate, retention sweeps from one toward zero and the continuation pressure climbs without bound.

Emb. day	Q (leaving)	T_c (h)	$x = A/(A+B)$	$\chi = [Q/(1-Q)]/x$
E_{11.0}	0.080	9.0	1.000	0.087
E_{11.4}	0.091	10.2	0.920	0.109
E_{11.8}	0.103	11.4	0.871	0.132
E_{12.2}	0.125	15.1	0.835	0.170
E_{12.6}	0.159	17.5	0.797	0.238
E_{13.1}	0.201	18.4	0.746	0.337

Emb. day	Q (leaving)	Tc (h)	$x = A/(A+B)$	$\chi = [Q/(1-Q)]/x$
E13.6	0.274	18.4	0.683	0.552
E14.1	0.370	18.4	0.589	0.996
E14.8	0.530	18.4	0.467	2.414
E15.5	0.567	18.4	0.299	4.370
E16.2	0.616	18.4	0.200	8.049
E17.0	0.892	18.4	0.128	64.961

4. THE BLIND SCREEN

The rule was frozen, with its SHA-256 recorded, before the trajectory was integrated. The bridge asks that the machine settle at its fixed point — retention $x = r$ and continuation $\chi = \sqrt{5}$ reached together, within the residual ceiling.

PREREGISTERED before the trajectory was integrated (16 July 2026):

Dataset = murine neocortical neuronogenesis, Takahashi, Nowakowski & Caviness (J Neurosci 1996):

leaving fraction Q per embryonic day (Table 3) and cell-cycle length Tc (Table 2), E11-E17, 11 cycles.

Whole = the accumulating proliferative epithelium over the neuronogenetic interval (a growing history, not a steady state).

Currency = cells; classes progenitor (retained) and neuron (terminal, monotonically accumulating); x = progenitors/(progenitors+neurons).

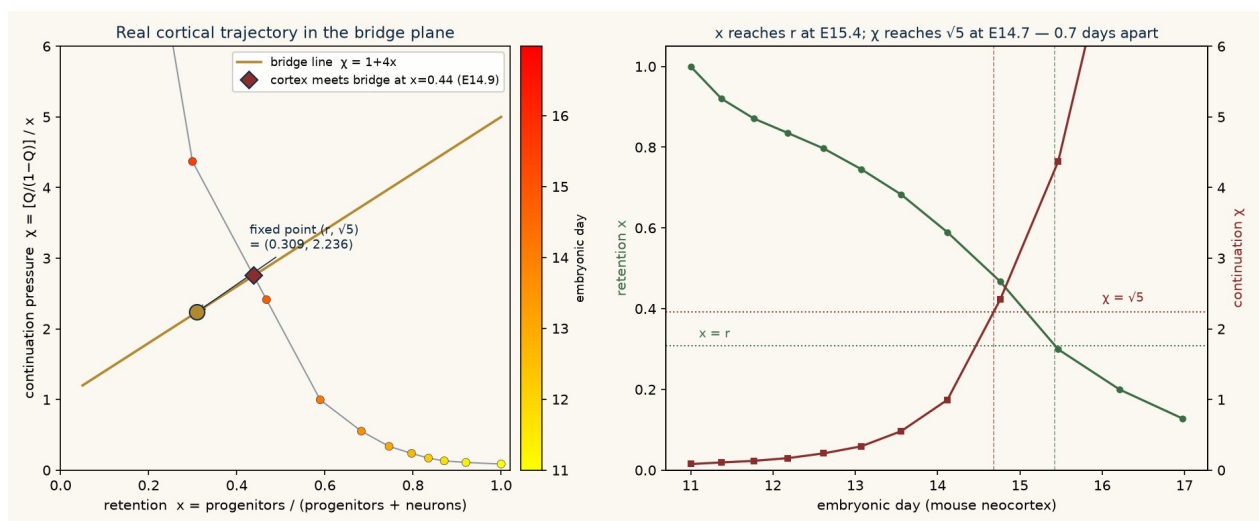
Mapping = per cycle $A \rightarrow 2(1-Q)A$, $B \rightarrow B + 2QA$; production $P = 2(1-Q)A$, conversion $K = 2QA$; $\chi = [Q/(1-Q)]/x$; $\beta = 0$.

Bridge = $\chi = 1 + 4x$; fixed point $x^* = r = 0.309017$, $\chi^* = 1 + 4r = \sqrt{5} = 2.236068$.

Frozen rule = PASS iff x settles at r and χ settles at $\sqrt{5}$ simultaneously, $|\chi - \sqrt{5}| \leq r^3 = 0.0295$ at the $x=r$ crossing.

No parameter is fitted; $Q(t)$ and $Tc(t)$ are the measured record.

SHA-256 of the frozen rule: 57ffad5f3c313c37f2a49c4a3c51e6ce7b9c5bd343095ce29eb32f072f4a7355



Left: the real cortical trajectory (points coloured by embryonic day) in the plane of retention against continuation pressure; the gold line is the bridge relation $\chi = 1 + 4x$ and the gold disc its fixed point $(r, \sqrt{5})$. The trajectory crosses the bridge line at $x = 0.44$ near E14.9, away from the fixed point. Right: retention reaches r at E15.4 while continuation reaches $\sqrt{5}$ at E14.7 — the two never coincide.

RECORD — REAL-DATA SCREEN, FAIL WITH A GRAZE — Retention passes through r at E15.4; continuation passes through $\sqrt{5}$ at E14.7; the crossings are 0.7 embryonic days apart, so the fixed point is never realised. At the $x = r$ crossing $\chi = 4.26$, missing $\sqrt{5}$ by 2.02, far outside the ceiling $r^3 = 0.0295$. The screen fails. It fails narrowly: the trajectory does cross the bridge relation $\chi = 1+4x$ once, at $x = 0.44$ near E14.9 — it grazes the bridge, but at the wrong retention.

5. WHAT THE GRAZE DIAGNOSES

This is a sharper, closer failure than the avalanche case, and on real measured data. The cortical machine is a ramp, not a plateau: the leaving fraction climbs continuously from zero to one, so retention sweeps through r rather than resting at it, and the continuation pressure sweeps through $\sqrt{5}$ at a different moment. The bridge is a statement about a fixed point — a value a self-regulating machine settles onto and holds — and a ramping developmental sweep, by construction, holds nothing. The trajectory grazing the bridge relation at E14.9 is the geometric signature of a sweep passing near a fixed point it does not occupy, exactly as the diagnosis of Application 02 anticipated.

DIAGNOSIS — SWEEP, NOT PLATEAU (CONFIRMED ON DATA) — Mammalian corticogenesis ramps its leaving fraction monotonically, so retention is never stationary and the bridge fixed point is not occupied. The screen correctly rejects it. The bridge's $\sqrt{5}$ lives in a system that holds its leaving fraction constant — a balanced self-renewing compartment at steady turnover, where retention settles at r and continuation at $\sqrt{5}$ together. That is the next dataset.

The remaining ambiguity is stated honestly. The extraction depends on the declared mapping — gross births into each class, $\beta = 0$ — and a different defensible accounting of production shifts the fixed-point leaving fraction. Fixing that mapping by the physics of the specific compartment, before any comparison, is a required preregistration step; it is also why the balanced-turnover test is stronger, since there the retained fraction is read directly from a standing progenitor-to-neuron ratio rather than reconstructed through an accumulation integral.

6. THE NEXT TEST, AND THE CONTROL

The preregistered target now moves to a compartment that holds its leaving fraction stationary. Adult and homeostatic neurogenesis — the subventricular and subgranular zones, or a self-renewing neural-stem-cell niche at balanced turnover — maintain a standing progenitor pool against continuous neuron output. There the retention coordinate is a directly measured standing ratio, $x = \text{progenitors}/(\text{progenitors} + \text{new neurons})$, and the frozen screen is a single measurement: does x sit at $r = 0.309$, equivalently does the steady leaving fraction hold the continuation pressure at $\sqrt{5}$, within r^3 . The invariant lineage of *C. elegans* provides the developmental control, the analogue of radioactive branching: a conserved two-class accumulation with a hard-wired programme and no feedback on the split, which the compiler should reject as a machine even where its terminal fraction is computable.

The standing result is a real-data failure that improves the question. Across three substrates now — stellar carbon-oxygen, cortical avalanches, and cortical neurogenesis — the compiler has made a falsifiable commitment away from cosmology and been told, each time with a specific diagnosis, either wrong regime or wrong Whole. That is the framework doing work: not interpreting everything, but isolating one machine at a time, screening it blind against a number fixed in advance, and recording

where and why it fails. The bridge's single empirical claim — continuation pressure $\sqrt{5}$ under genuine self-regulation — remains open and now has its sharpest test named: a standing progenitor-to-neuron ratio of r in a balanced neurogenic niche.